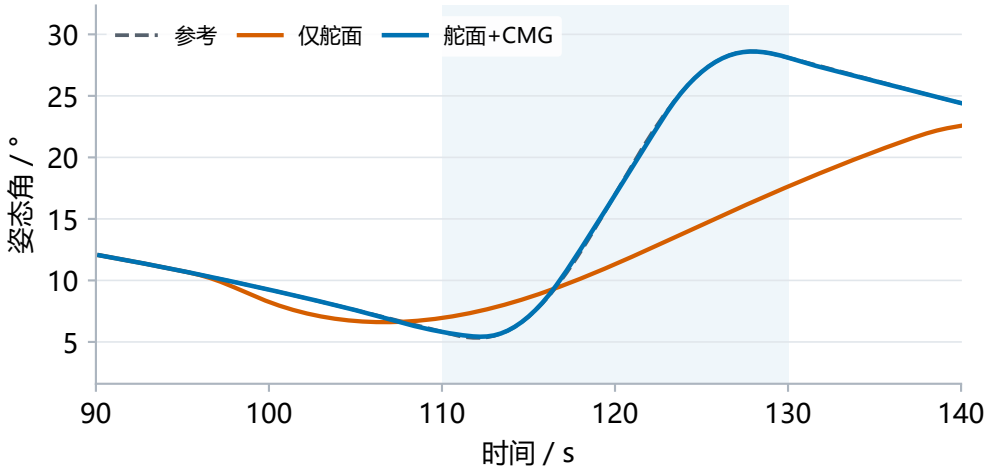
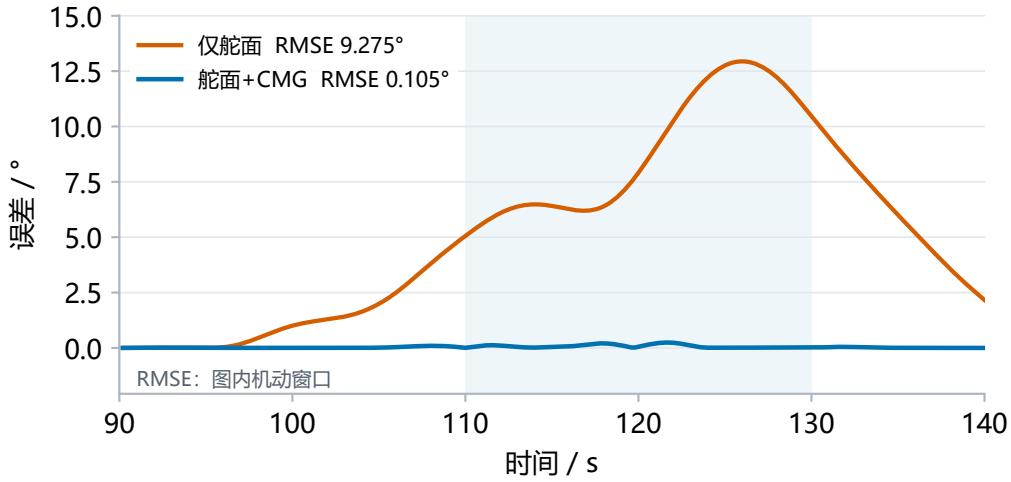
P1 正向30°俯仰：舵面与CMG协同控制

显示90–140 s；浅蓝背景：机动窗口；浅绿区域及绿虚线：气动力矩可行区间与上下限

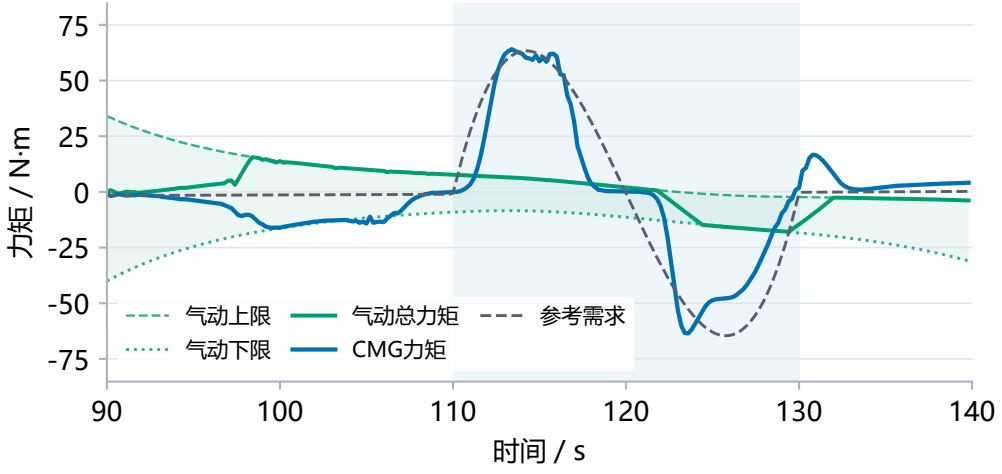
(a) 俯仰姿态跟踪



(b) 三轴姿态误差范数



(c) 联合控制的俯仰轴力矩分工



(d) CMG俯仰轴角动量

